Greenpod Labs — Assignment 2

Computer Vision: Fruit Condition Classification

1. Introduction

The objective of this project was to develop and compare two computer vision approaches for image classification. The experiment implemented a classical machine learning model based on color and texture features and a deep learning model using MobileNetV2 transfer learning.

The available dataset contained orange-condition classes: blackspot, canker, fresh, and grenning. The original assignment specifies avocado ripening-stage classification; therefore, the present results represent an orange-condition classification experiment rather than a completed avocado classification solution.

2. Dataset Understanding

The dataset contained 1,090 images in total, consisting of 991 training images and 99 test images.

The four classes were:

- Blackspot
- Canker
- Fresh
- Grenning

The test-set distribution was:

Class	Number of test images
Blackspot	22
Canker	22
Fresh	33
Grenning	22
Total	**99**

The test set is not perfectly balanced because the fresh class has 33 images, while each other class has 22. This difference supports reporting macro precision, macro recall, and macro F1-score in addition to accuracy.

Potential sources of bias include changes in lighting, shadows, image backgrounds, camera angles, resolution, and variation in the appearance of fruit symptoms. Images from the same original source or near-duplicate images appearing in both training and test sets could also cause data leakage. These possibilities should be checked during further data analysis.

3. Data Preprocessing

Images were resized to a fixed dimension before feature extraction or deep learning. The classical approach used 128 × 128 images, while MobileNetV2 used 224 × 224 images.

For the classical approach, color information was represented using HSV histograms and texture information was represented using Local Binary Patterns. For MobileNetV2, images were normalized using the ImageNet mean and standard deviation, and training images were augmented using horizontal flips, small rotations, and mild brightness and contrast changes.

4. Approach 1: Classical Computer Vision and SVM

The first approach combined handcrafted image features with a Support Vector Machine classifier.

HSV histograms represented the distribution of hue, saturation, and brightness in each image. Local Binary Patterns represented local texture patterns. The features were concatenated and standardized before training an SVM with a radial basis function kernel.

This approach was selected because it provides a computationally practical baseline for a relatively small dataset. It is also useful for understanding how color and texture features contribute to classification.

The SVM achieved **97.98% test accuracy** and a **97.73% macro F1-score**.

5. Approach 2: MobileNetV2 Transfer Learning

The second approach used MobileNetV2 pretrained on ImageNet. The feature extraction layers were frozen, and the final classification layer was replaced with a four-class output layer.

Transfer learning was selected because pretrained convolutional layers can capture general visual patterns such as edges, textures, and shapes. MobileNetV2 is also designed to use fewer computational resources than many larger neural network architectures.

The model was trained for three epochs using cross-entropy loss and the Adam optimizer.

MobileNetV2 achieved **93.94% test accuracy** and a **93.42% macro F1-score**.

6. Evaluation Metrics and Model Comparison

Accuracy measures the proportion of test images classified correctly. Precision measures how many predictions assigned to a class were correct. Recall measures how many actual samples of a class were identified. F1-score combines precision and recall.

Macro averages give each class equal weight, which is useful when class frequencies differ.

Metric	SVM	MobileNetV2
Accuracy	97.98%	93.94%
Macro precision	97.73%	93.80%
Macro recall	97.73%	93.18%

Metric	SVM	MobileNetV2
Macro F1-score	97.73%	93.42%

Based on the measured results, SVM performed better than MobileNetV2 on this test set. It achieved higher accuracy and higher macro precision, recall, and F1-score.

This result does not establish that SVM will always outperform deep learning. The test set contains only 99 images, and additional evaluation on a larger, independent dataset is needed before drawing conclusions about real-world performance.

7. Example Run Demonstration

A test image located in the canker class was passed through both trained models.

- **SVM prediction:** Canker
- **MobileNetV2 prediction:** Canker
- **MobileNetV2 softmax score:** 54.44%

Both models agreed on the predicted class for this example. The softmax score represents the model's relative output score and is not necessarily a calibrated probability of correctness.

8. Mandatory End-of-Assignment Questions and Answers

Question 1: In production, how does your model perform?

In the current experiment, the SVM achieved 97.98% test accuracy, while MobileNetV2 achieved 93.94% test accuracy on 99 test images. These results indicate that both approaches classified the available test images effectively, with SVM performing better on the measured metrics.

However, these figures are offline test results and should not be interpreted as guaranteed production performance. Actual production performance depends on camera quality, lighting, background variation, fruit varieties, processing hardware, and the similarity between deployment images and training data.

Inference speed must also be measured on the intended hardware. GPU and CPU latency values should not be reported without conducting and documenting actual benchmarks. In a sorting line, latency and throughput should be measured under realistic operating conditions.

Question 2: What are the drawbacks of your model?

The main limitations are:

1. **Dependence on visible appearance:** Color and texture features may be affected by lighting, shadows, reflections, and background variation.
2. **Overlapping classes:** Different fruit conditions may have similar visible patterns, making some examples difficult to distinguish.
3. **Limited dataset size:** The test set contains only 99 images, so its metrics may not represent performance across all real-world conditions.

4. **Potential dataset bias:** The model may learn correlations with the backgrounds, image sources, or acquisition conditions instead of relying entirely on fruit characteristics.
5. **Model confidence:** MobileNetV2's softmax score may not be calibrated, so a high score does not guarantee a correct prediction.
6. **Task limitation:** The current models classify orange conditions and do not determine internal fruit quality or avocado ripening stages.

Question 3: If given more time, what would you improve?

I would improve the project in the following ways:

1. **Collect more diverse data:** Add independently collected images covering different fruit varieties, cameras, backgrounds, and lighting conditions.
2. **Improve dataset validation:** Check incorrect labels, corrupted images, class imbalance, and duplicate or near-duplicate images across data splits.
3. **Use a validation set:** Tune hyperparameters and select models using a validation set, keeping the test set separate for final evaluation.
4. **Improve deployment performance:** Measure inference latency and throughput on the intended hardware, then investigate ONNX export, quantization, or other optimizations.
5. **Improve robustness:** Evaluate performance under shadows, glare, changes in image exposure, and other realistic environmental conditions.
6. **Use the specified avocado dataset:** For the original assignment, obtain the avocado dataset and retrain both approaches to classify underripe, breaking, ripe, and overripe stages.
7. **Explore additional sensing:** If the application requires estimating internal ripeness, investigate non-destructive near-infrared sensing or firmness measurements alongside visual classification.

Question 4: Where did you spend the most time while solving your assignment?

The main areas of work were:

1. **Dataset inspection and preprocessing:** Understanding the folder structure, checking the available classes, loading images, resizing them, and preparing the data for training.
2. **Feature engineering:** Extracting HSV color histograms and Local Binary Pattern texture features and preparing them for SVM classification.
3. **Model training:** Implementing the SVM pipeline and configuring MobileNetV2 transfer learning.
4. **Evaluation and comparison:** Calculating accuracy, precision, recall, and F1-score and examining the classification reports and confusion matrices.
5. **Demonstration and reporting:** Saving the models, testing both approaches on a sample image, and organizing the results for documentation.

The work involved all these areas. Exact time percentages were not tracked, so assigning numerical percentages would not be reliable.

9. Conclusion

Two computer vision approaches were implemented and compared on the available orange-condition dataset. The classical SVM model achieved 97.98% test accuracy and a 97.73% macro F1-score. MobileNetV2 achieved 93.94% test accuracy and a 93.42% macro F1-score.

SVM performed better on the current test set, while MobileNetV2 provides a transferable deep learning approach that could be investigated further with more data and additional tuning.

The current findings are limited to orange-condition classification. To satisfy the original assignment as written, the workflow must be repeated with the specified avocado dataset and evaluated on avocado ripening-stage labels.